Single PDF RAG Assistant Detailed CV Entry

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Project entry for a technical CV. Focus: applied generative AI, information retrieval and evidence-based evaluation.

Objective and implementation

Built a local PDF assistant that retrieves source evidence and generates answers with page citations using Streamlit, MiniLM, FAISS and Gemini Flash-Lite.

- Extracted page text and created 120-word windows with 25-word overlap while retaining page metadata.
- Computed normalized 384-dimensional MiniLM vectors on CPU and persisted an exact FAISS inner-product index with JSON chunk metadata.
- Used PDF-content and configuration hashes to reuse saved indexes for unchanged documents.
- Combined semantic candidates with term-rarity-weighted keyword candidates, expanded same-page neighbors and removed duplicate chunks.
- Integrated the Google generation SDK with a request timeout, one attempt and visible error reporting.
- Exposed retrieved passages to support manual assessment of answers and page citations.

Skills demonstrated

Python application development; NLP embeddings; vector search; document preprocessing; cache design; API integration; retrieval diagnostics; manual evaluation.

Evaluation and debugging evidence

A manually reviewed baseline passed 11 of 12 questions on the Transformer paper. The hardware-and-schedule question falsely reported missing information.

Inspection of the exact prompt showed pages 5, 8 and 9 but no page 7. This isolated a retrieval coverage failure. Hybrid retrieval and neighbor expansion recovered the hardware passage and produced a correct answer with page references.

The developer subsequently reported 12 of 12 passes in a full retest of the same development/regression set, including three missing-information questions. Complete final logs were not supplied for independent reproduction.

Engineering decisions

Moved from cloud embeddings to local MiniLM after an embedding-quota interruption. This removed embedding API quota usage while retaining the external Gemini generation dependency. Saved indexes avoid re-embedding unchanged PDFs after restarting.

Scope and next work

Local educational app; no production-scale deployment, business impact or general accuracy claim. Next steps are an exact-source retest, unseen-document evaluation, bounded context selection, stronger citation checks and eventual OCR or multi-PDF support.

GitHub: https://github.com/Get-AZ/Getamesay-Professional-Portfolio/tree/main/projects/Single-PDF-RAG-Assistant